

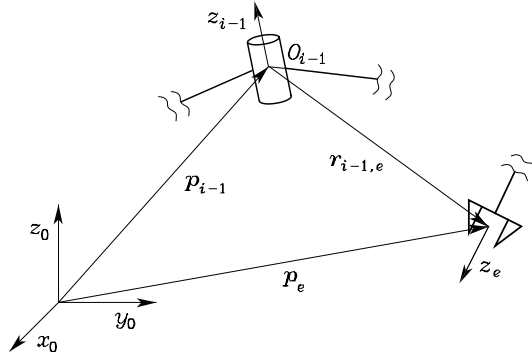


Fig. 3.2. Representation of vectors needed for the computation of the velocity contribution of a revolute joint to the end-effector linear velocity

3.1.3 Jacobian Computation

In order to compute the Jacobian, it is convenient to proceed separately for the linear velocity and the angular velocity.

For the contribution to the *linear velocity*, the time derivative of $\mathbf{p}_e(\mathbf{q})$ can be written as

$$\dot{\mathbf{p}}_e = \sum_{i=1}^n \frac{\partial \mathbf{p}_e}{\partial q_i} \dot{q}_i = \sum_{i=1}^n \mathbf{J}_{Pi} \dot{q}_i. \quad (3.27)$$

This expression shows how $\dot{\mathbf{p}}_e$ can be obtained as the sum of the terms $\dot{q}_i \mathbf{J}_{Pi}$. Each term represents the contribution of the velocity of single Joint i to the end-effector linear velocity when all the other joints are still.

Therefore, by distinguishing the case of a *prismatic* joint ($q_i = d_i$) from the case of a *revolute* joint ($q_i = \theta_i$), it is:

- If Joint i is *prismatic*, from (3.20) it is

$$\dot{q}_i \mathbf{J}_{Pi} = \dot{d}_i \mathbf{z}_{i-1}$$

and then

$$\mathbf{J}_{Pi} = \mathbf{z}_{i-1}.$$

- If Joint i is *revolute*, observing that the contribution to the linear velocity is to be computed with reference to the origin of the end-effector frame (Fig. 3.2), it is

$$\dot{q}_i \mathbf{J}_{Pi} = \boldsymbol{\omega}_{i-1,i} \times \mathbf{r}_{i-1,e} = \dot{\theta}_i \mathbf{z}_{i-1} \times (\mathbf{p}_e - \mathbf{p}_{i-1})$$

and then

$$\mathbf{J}_{Pi} = \mathbf{z}_{i-1} \times (\mathbf{p}_e - \mathbf{p}_{i-1}).$$

For the contribution to the *angular velocity*, in view of (3.18), it is

$$\boldsymbol{\omega}_e = \boldsymbol{\omega}_n = \sum_{i=1}^n \boldsymbol{\omega}_{i-1,i} = \sum_{i=1}^n \mathbf{J}_{Oi} \dot{q}_i, \quad (3.28)$$

where (3.19) and (3.23) have been utilized to characterize the terms $\dot{q}_i \mathbf{J}_{Oi}$, and thus in detail:

- If Joint i is *prismatic*, from (3.19) it is

$$\dot{q}_i \mathbf{J}_{Oi} = \mathbf{0}$$

and then

$$\mathbf{J}_{Oi} = \mathbf{0}.$$

- If Joint i is *revolute*, from (3.23) it is

$$\dot{q}_i \mathbf{J}_{Oi} = \dot{\vartheta}_i \mathbf{z}_{i-1}$$

and then

$$\mathbf{J}_{Oi} = \mathbf{z}_{i-1}.$$

In summary, the Jacobian in (3.5) can be partitioned into the (3×1) column vectors $\mathbf{J}_{Pi}$ and $\mathbf{J}_{Oi}$ as

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_{P1} & \cdots & \mathbf{J}_{Pn} \\ \mathbf{J}_{O1} & & \mathbf{J}_{On} \end{bmatrix}, \quad (3.29)$$

where

$$\begin{bmatrix} \mathbf{J}_{Pi} \\ \mathbf{J}_{Oi} \end{bmatrix} = \begin{cases} \begin{bmatrix} \mathbf{z}_{i-1} \\ \mathbf{0} \end{bmatrix} & \text{for a } \textit{prismatic} \text{ joint} \\ \begin{bmatrix} \mathbf{z}_{i-1} \times (\mathbf{p}_e - \mathbf{p}_{i-1}) \\ \mathbf{z}_{i-1} \end{bmatrix} & \text{for a } \textit{revolute} \text{ joint.} \end{cases} \quad (3.30)$$

The expressions in (3.30) allow Jacobian computation in a simple, systematic way on the basis of direct kinematics relations. In fact, the vectors $\mathbf{z}_{i-1}$, $\mathbf{p}_e$ and $\mathbf{p}_{i-1}$ are all functions of the joint variables. In particular:

- $\mathbf{z}_{i-1}$ is given by the third column of the rotation matrix $\mathbf{R}_{i-1}^0$, i.e.,

$$\mathbf{z}_{i-1} = \mathbf{R}_1^0(q_1) \cdots \mathbf{R}_{i-1}^{i-2}(q_{i-1}) \mathbf{z}_0 \quad (3.31)$$

where $\mathbf{z}_0 = [0 \ 0 \ 1]^T$ allows the selection of the third column.

- $\mathbf{p}_e$ is given by the first three elements of the fourth column of the transformation matrix $\mathbf{T}_e^0$, i.e., by expressing $\tilde{\mathbf{p}}_e$ in the (4×1) homogeneous form

$$\tilde{\mathbf{p}}_e = \mathbf{A}_1^0(q_1) \cdots \mathbf{A}_n^{n-1}(q_n) \tilde{\mathbf{p}}_0 \quad (3.32)$$

where $\tilde{\mathbf{p}}_0 = [0 \ 0 \ 0 \ 1]^T$ allows the selection of the fourth column.

- $\mathbf{p}_{i-1}$ is given by the first three elements of the fourth column of the transformation matrix $\mathbf{T}_{i-1}^0$, i.e., it can be extracted from

$$\tilde{\mathbf{p}}_{i-1} = \mathbf{A}_1^0(q_1) \dots \mathbf{A}_{i-1}^{i-2}(q_{i-1}) \tilde{\mathbf{p}}_0. \quad (3.33)$$

The above equations can be conveniently used to compute the translational and rotational velocities of any point along the manipulator structure, as long as the direct kinematics functions relative to that point are known.

Finally, notice that the Jacobian matrix depends on the frame in which the end-effector velocity is expressed. The above equations allow computation of the geometric Jacobian with respect to the base frame. If it is desired to represent the Jacobian in a different Frame u , it is sufficient to know the relative rotation matrix $\mathbf{R}^u$. The relationship between velocities in the two frames is

$$\begin{bmatrix} \dot{\mathbf{p}}_e^u \\ \boldsymbol{\omega}_e^u \end{bmatrix} = \begin{bmatrix} \mathbf{R}^u & \mathbf{O} \\ \mathbf{O} & \mathbf{R}^u \end{bmatrix} \begin{bmatrix} \dot{\mathbf{p}}_e \\ \boldsymbol{\omega}_e \end{bmatrix},$$

which, substituted in (3.4), gives

$$\begin{bmatrix} \dot{\mathbf{p}}_e^u \\ \boldsymbol{\omega}_e^u \end{bmatrix} = \begin{bmatrix} \mathbf{R}^u & \mathbf{O} \\ \mathbf{O} & \mathbf{R}^u \end{bmatrix} \mathbf{J} \dot{\mathbf{q}}$$

and then

$$\mathbf{J}^u = \begin{bmatrix} \mathbf{R}^u & \mathbf{O} \\ \mathbf{O} & \mathbf{R}^u \end{bmatrix} \mathbf{J}, \quad (3.34)$$

where $\mathbf{J}^u$ denotes the geometric Jacobian in Frame u , which has been assumed to be time-invariant.

3.2 Jacobian of Typical Manipulator Structures

In the following, the Jacobian is computed for some of the typical manipulator structures presented in the previous chapter.

3.2.1 Three-link Planar Arm

In this case, from (3.30) the Jacobian is

$$\mathbf{J}(\mathbf{q}) = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{p}_3 - \mathbf{p}_0) & \mathbf{z}_1 \times (\mathbf{p}_3 - \mathbf{p}_1) & \mathbf{z}_2 \times (\mathbf{p}_3 - \mathbf{p}_2) \\ \mathbf{z}_0 & \mathbf{z}_1 & \mathbf{z}_2 \end{bmatrix}.$$

Computation of the position vectors of the various links gives

$$\mathbf{p}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{p}_1 = \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ 0 \end{bmatrix} \quad \mathbf{p}_2 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} \\ a_1 s_1 + a_2 s_{12} \\ 0 \end{bmatrix}$$

$$\mathbf{p}_3 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ 0 \end{bmatrix}$$

while computation of the unit vectors of revolute joint axes gives

$$\mathbf{z}_0 = \mathbf{z}_1 = \mathbf{z}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

since they are all parallel to axis z_0 . From (3.29) it is

$$\mathbf{J} = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} - a_3 s_{123} & -a_2 s_{12} - a_3 s_{123} & -a_3 s_{123} \\ a_1 c_1 + a_2 c_{12} + a_3 c_{123} & a_2 c_{12} + a_3 c_{123} & a_3 c_{123} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}. \quad (3.35)$$

In the Jacobian (3.35), only the three non-null rows are relevant (the rank of the matrix is at most 3); these refer to the two components of linear velocity along axes x_0 , y_0 and the component of angular velocity about axis z_0 . This result can be derived by observing that three DOFs allow specification of at most three end-effector variables; v_z , ω_x , ω_y are always null for this kinematic structure. If orientation is of no concern, the (2×3) Jacobian for the positional part can be derived by considering just the first two rows, i.e.,

$$\mathbf{J}_P = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} - a_3 s_{123} & -a_2 s_{12} - a_3 s_{123} & -a_3 s_{123} \\ a_1 c_1 + a_2 c_{12} + a_3 c_{123} & a_2 c_{12} + a_3 c_{123} & a_3 c_{123} \end{bmatrix}. \quad (3.36)$$

3.2.2 Anthropomorphic Arm

In this case, from (3.30) the Jacobian is

$$\mathbf{J} = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{p}_3 - \mathbf{p}_0) & \mathbf{z}_1 \times (\mathbf{p}_3 - \mathbf{p}_1) & \mathbf{z}_2 \times (\mathbf{p}_3 - \mathbf{p}_2) \\ \mathbf{z}_0 & \mathbf{z}_1 & \mathbf{z}_2 \end{bmatrix}.$$

Computation of the position vectors of the various links gives

$$\mathbf{p}_0 = \mathbf{p}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{p}_2 = \begin{bmatrix} a_2 c_1 c_2 \\ a_2 s_1 c_2 \\ a_2 s_2 \end{bmatrix}$$

$$\mathbf{p}_3 = \begin{bmatrix} c_1(a_2 c_2 + a_3 c_{23}) \\ s_1(a_2 c_2 + a_3 c_{23}) \\ a_2 s_2 + a_3 s_{23} \end{bmatrix}$$

while computation of the unit vectors of revolute joint axes gives

$$\mathbf{z}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{z}_1 = \mathbf{z}_2 = \begin{bmatrix} s_1 \\ -c_1 \\ 0 \end{bmatrix}.$$

From (3.29) it is

$$\mathbf{J} = \begin{bmatrix} -s_1(a_2c_2 + a_3c_{23}) & -c_1(a_2s_2 + a_3s_{23}) & -a_3c_1s_{23} \\ c_1(a_2c_2 + a_3c_{23}) & -s_1(a_2s_2 + a_3s_{23}) & -a_3s_1s_{23} \\ 0 & a_2c_2 + a_3c_{23} & a_3c_{23} \\ 0 & s_1 & s_1 \\ 0 & -c_1 & -c_1 \\ 1 & 0 & 0 \end{bmatrix}. \quad (3.37)$$

Only three of the six rows of the Jacobian (3.37) are linearly independent. Having 3 DOFs only, it is worth considering the upper (3×3) block of the Jacobian

$$\mathbf{J}_P = \begin{bmatrix} -s_1(a_2c_2 + a_3c_{23}) & -c_1(a_2s_2 + a_3s_{23}) & -a_3c_1s_{23} \\ c_1(a_2c_2 + a_3c_{23}) & -s_1(a_2s_2 + a_3s_{23}) & -a_3s_1s_{23} \\ 0 & a_2c_2 + a_3c_{23} & a_3c_{23} \end{bmatrix} \quad (3.38)$$

that describes the relationship between the joint velocities and the end-effector linear velocity. This structure does not allow an arbitrary angular velocity $\boldsymbol{\omega}$ to be obtained; in fact, the two components ω_x and ω_y are not independent ($s_1\omega_y = -c_1\omega_x$).

3.2.3 Stanford Manipulator

In this case, from (3.30) it is

$$\mathbf{J} = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{p}_6 - \mathbf{p}_0) & \mathbf{z}_1 \times (\mathbf{p}_6 - \mathbf{p}_1) & \mathbf{z}_2 \\ \mathbf{z}_0 & \mathbf{z}_1 & \mathbf{0} \\ \mathbf{z}_3 \times (\mathbf{p}_6 - \mathbf{p}_3) & \mathbf{z}_4 \times (\mathbf{p}_6 - \mathbf{p}_4) & \mathbf{z}_5 \times (\mathbf{p}_6 - \mathbf{p}_5) \\ \mathbf{z}_3 & \mathbf{z}_4 & \mathbf{z}_5 \end{bmatrix}.$$

Computation of the position vectors of the various links gives

$$\mathbf{p}_0 = \mathbf{p}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{p}_3 = \mathbf{p}_4 = \mathbf{p}_5 = \begin{bmatrix} c_1s_2d_3 - s_1d_2 \\ s_1s_2d_3 + c_1d_2 \\ c_2d_3 \end{bmatrix}$$

$$\mathbf{p}_6 = \begin{bmatrix} c_1s_2d_3 - s_1d_2 + (c_1(c_2c_4s_5 + s_2c_5) - s_1s_4s_5)d_6 \\ s_1s_2d_3 + c_1d_2 + (s_1(c_2c_4s_5 + s_2c_5) + c_1s_4s_5)d_6 \\ c_2d_3 + (-s_2c_4s_5 + c_2c_5)d_6 \end{bmatrix},$$

while computation of the unit vectors of joint axes gives

$$\begin{aligned} \mathbf{z}_0 &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & \mathbf{z}_1 &= \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} & \mathbf{z}_2 = \mathbf{z}_3 &= \begin{bmatrix} c_1 s_2 \\ s_1 s_2 \\ c_2 \end{bmatrix} \\ \mathbf{z}_4 &= \begin{bmatrix} -c_1 c_2 s_4 - s_1 c_4 \\ -s_1 c_2 s_4 + c_1 c_4 \\ s_2 s_4 \end{bmatrix} & \mathbf{z}_5 &= \begin{bmatrix} c_1 (c_2 c_4 s_5 + s_2 c_5) - s_1 s_4 s_5 \\ s_1 (c_2 c_4 s_5 + s_2 c_5) + c_1 s_4 s_5 \\ -s_2 c_4 s_5 + c_2 c_5 \end{bmatrix}. \end{aligned}$$

The sought Jacobian can be obtained by developing the computations as in (3.29), leading to expressing end-effector linear and angular velocity as a function of joint velocities.

3.3 Kinematic Singularities

The Jacobian in the differential kinematics equation of a manipulator defines a linear mapping

$$\mathbf{v}_e = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}} \quad (3.39)$$

between the vector $\dot{\mathbf{q}}$ of joint velocities and the vector $\mathbf{v}_e = [\dot{\mathbf{p}}_e^T \ \boldsymbol{\omega}_e^T]^T$ of end-effector velocity. The Jacobian is, in general, a function of the configuration $\mathbf{q}$; those configurations at which $\mathbf{J}$ is rank-deficient are termed *kinematic singularities*. To find the singularities of a manipulator is of great interest for the following reasons:

- a) Singularities represent configurations at which mobility of the structure is reduced, i.e., it is not possible to impose an arbitrary motion to the end-effector.
- b) When the structure is at a singularity, infinite solutions to the inverse kinematics problem may exist.
- c) In the neighbourhood of a singularity, small velocities in the operational space may cause large velocities in the joint space.

Singularities can be classified into:

- *Boundary* singularities that occur when the manipulator is either out-stretched or retracted. It may be understood that these singularities do not represent a true drawback, since they can be avoided on condition that the manipulator is not driven to the boundaries of its reachable workspace.
- *Internal* singularities that occur inside the reachable workspace and are generally caused by the alignment of two or more axes of motion, or else by the attainment of particular end-effector configurations. Unlike the above, these singularities constitute a serious problem, as they can be encountered anywhere in the reachable workspace for a planned path in the operational space.